Driving workforce transition in Australian life science research

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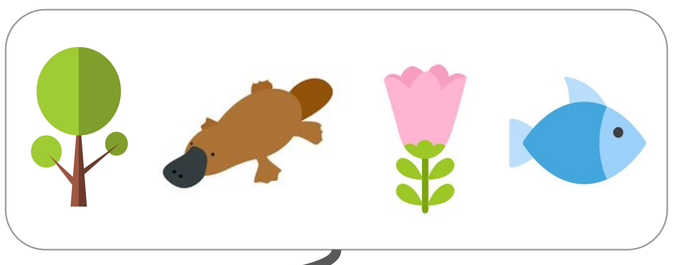
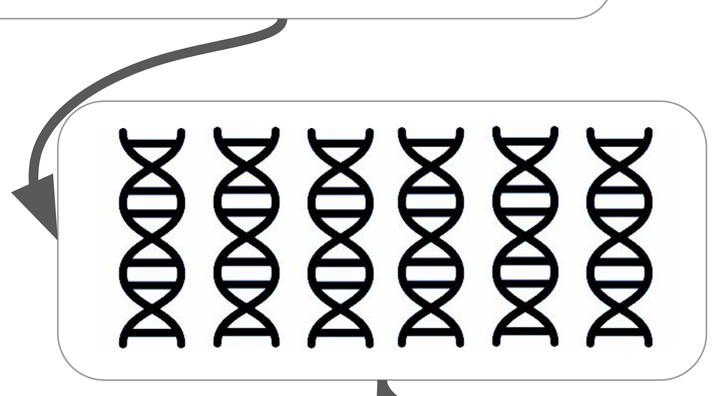
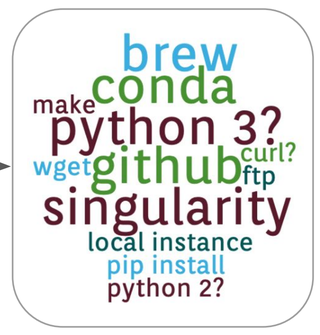
Genome sequencing projects are a current focus for the Australian life science research and conservation communities. These ambitious projects are being enabled through increasingly cheaper access to long-read sequencing technologies, which in turn are driving the need for greater computing power. The Australian BioCommons (https://www.biocommons.org.au/) is therefore working to build capacity in the researcher workforce by providing better access to analysis platforms and skills development. Here we present the process and challenges involved in configuring Galaxy Australia (https://usegalaxy.org.au/) for analyses of large genomic data sets, resulting in a national service capable of providing all the necessary tools for Australian researchers to best use the rich data derived from genome sequencing projects.



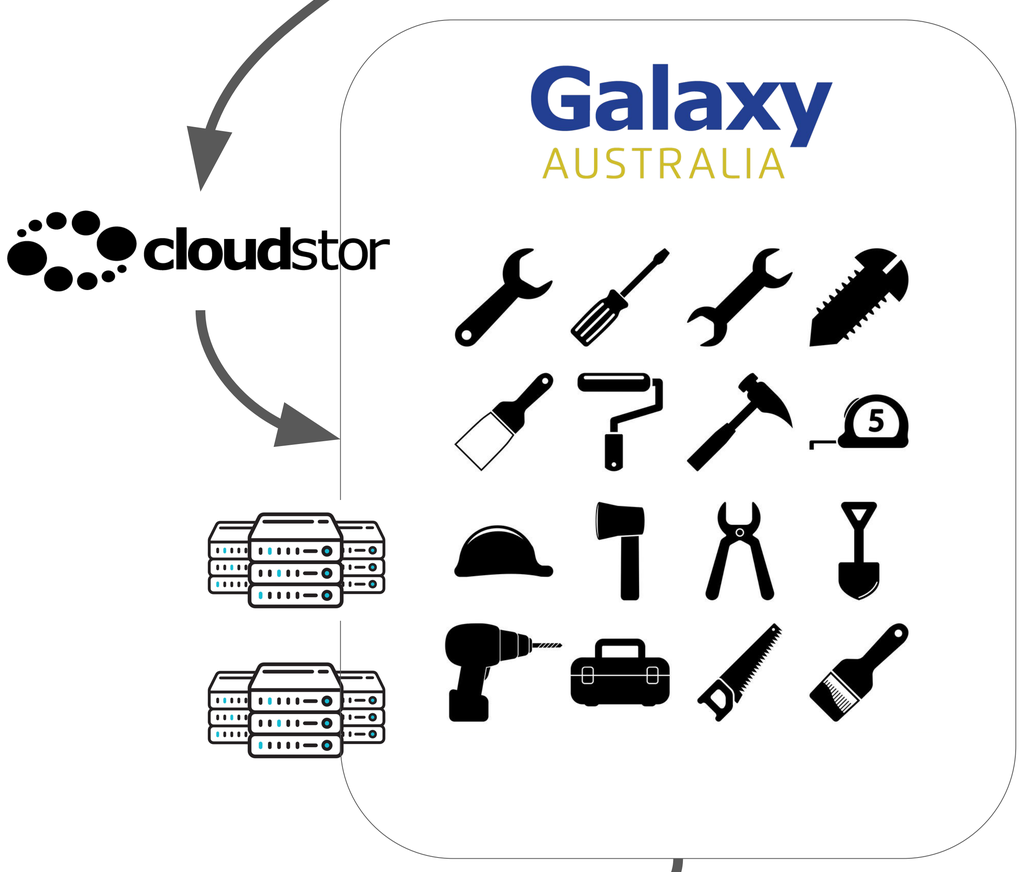
A current focus in Australian research is the generation of whole genome sequences for iconic flora and fauna. National consortia have been established to sequence complete genomes of plants, marsupials, threatened species, amphibians and reptiles, including threatened species (for example, https://bioplatforms.com/projects/).

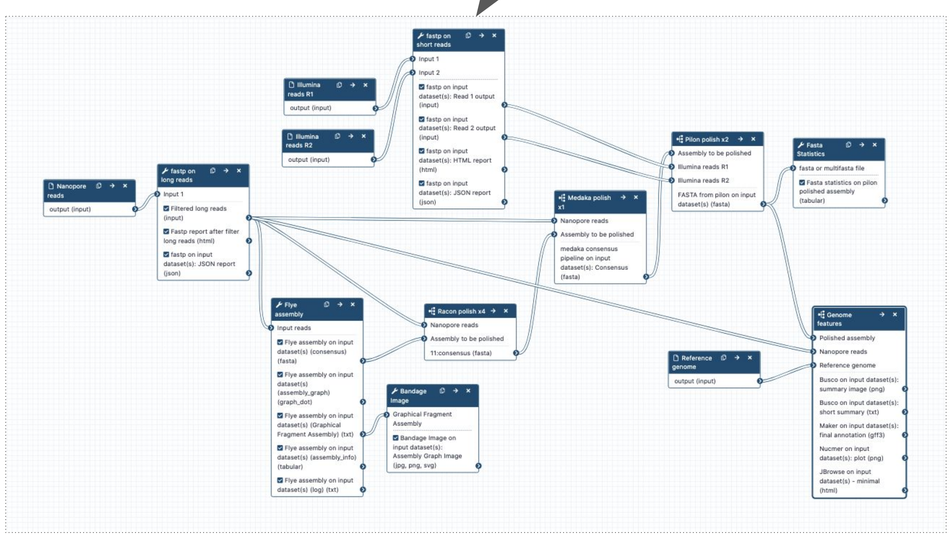


Large genome assembly is computationally-intensive, and there is high demand for the resources and associated bioinformatics skills to successfully complete these projects. Big data sets and big data analysis are the new normal, and scientists need access to tools, skills and computational resources to participate fully in the current research landscape.



With much of the data being generated by specialist sequencing providers, the first hurdle for researchers is often the biggest: ***where can I analyse my data; what tools can I use, and how should I string those tools together in a sensible workflow?***



Galaxy Australia is one option for these analyses, and the Australian BioCommons is working to expand its capacity and capability to support large genome assembly:

- First, CloudStor file storage has been connected to Galaxy Australia, so that researchers can easily import large files (https://www.aarnet.edu.au/network-and-services/cloud-services/cloudstor).
- Second, underlying high memory (>1TB) compute nodes have been installed and linked to particular tools, to facilitate computationally-intensive processes such as large genome assembly.
- Third, genome assembly tools have been tested by the Galaxy Australia team. This has has flow-on effects in improving the platform, as the testing has resulted in bug fixes and tool wrapper improvements.
- Fourth, new relevant tools have been installed. Additional tools surrounding the genome assembly analysis process have also been tested, such as those for quality control, polishing, and visualization.
- Fifth, workflows and accompanying tutorials are being developed for genome assembly, and tested on datasets of varying size and genomic complexity. Researchers can import workflows and customise as needed to run on their own data.
- In summary: We anticipate that this project will significantly improve the access and options for life scientists to better analyse their large genomic data sets. A constant challenge is staying current with continuous improvements in sequencing and computational technologies, and associated tools and research questions.